xAI 2: Causal Interpretability

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Today's lecture

- What is a causal explanation? How to set up a counterfactual test?
- Causal & Counterfactual interpretability methods

Learning Goals

Be able to:

- explain the difference between descriptive and causal interpretability methods and their explanatory power
- list potential locations for causal mediation/intervention in a transformer-based neural network model
- describe how counterfactual/causal interpretability methods (e.g. activation patching, steering) work

Awaken your inner child: demand an explanation

In pairs: Questioner & Explainer

Questioner:

- Toddler: "Why?" https://en.wikipedia.org/wiki/Five_whys
- Teenager: "How do you know?"

Ask & explain something, for example:

- Why am I here this morning?
- Why is the sky blue?
- Why do leaves fall in autumn?
- Why did our linear probe only get 45% accuracy?

Then switch! 4 minutes

Reflection

What makes a good explanation?

What makes a good explanation?

Mueller et al., 2026: The Quest for the Right Mediator: Surveying Mechanistic Interpretability for NLP Through the Lens of Causal Mediation Analysis

- **faithfulness** to model / reality
- **selectivity**: minimal explanation of a single phenomenon at a time
- **sparsity**: doesn't involve more of the model than necessary
- **generality**: applies to as much of the data space as possible

Explanations should be useful for communication:

- human **comprehensibility**
- **plausibility** to humans

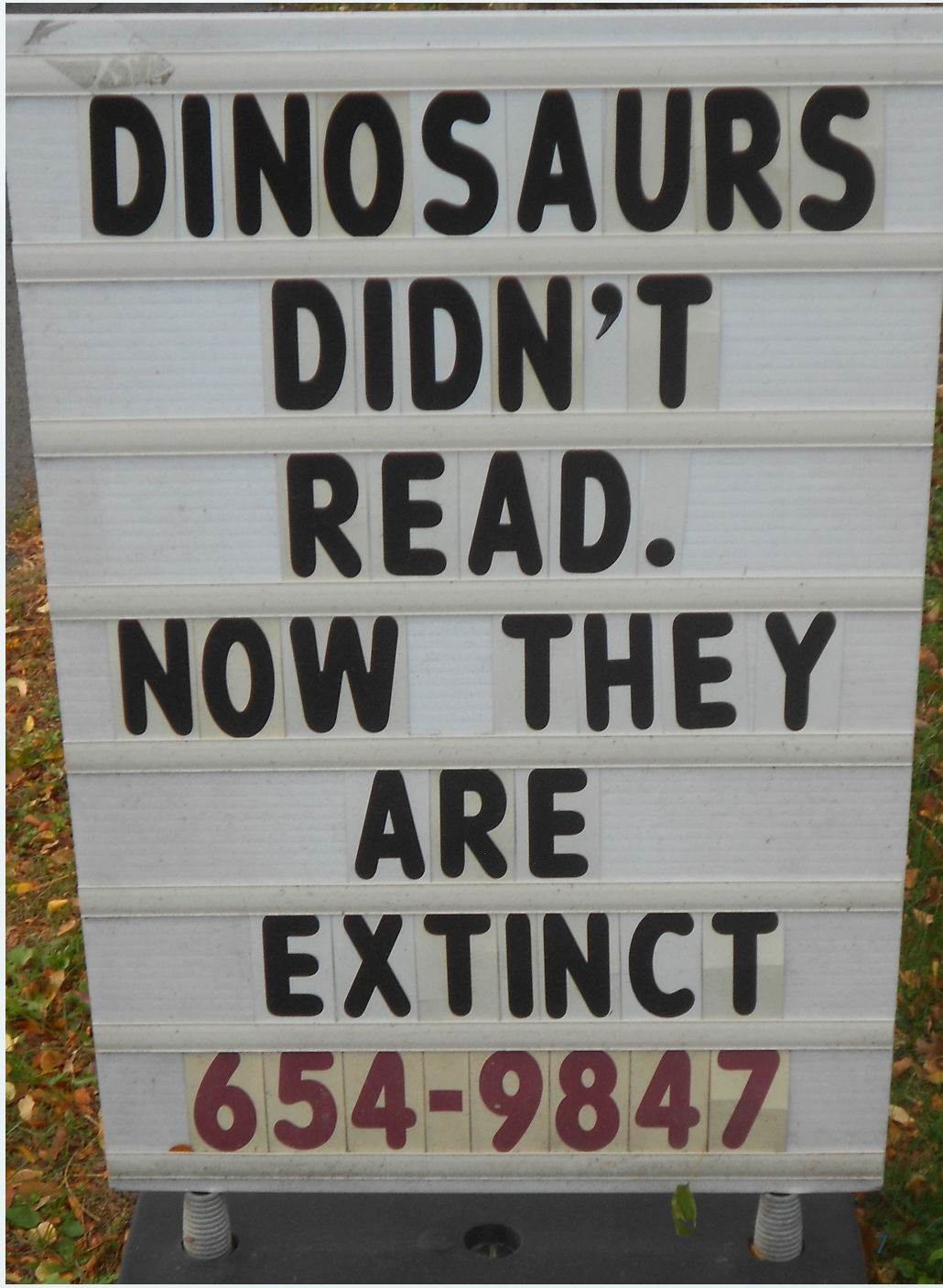
Causal interpretability

Last week was **descriptive** interpretability - exploratory analysis of model behaviours and representations

- e.g. neighbors, linear probing, attention flows

Today: **Causal** explanations & interpretability methods.

- What made the model do X?
- How can we find the causal mechanisms within a model?



The Counterfactual theory of Causality

> “An event E **causally depends** on C [iff]
> (i) if C had occurred, then E would have occurred, and
> (ii) if C had not occurred, then E would not have occurred.” - David Lewis (1973)

Counterfactual: "if not-C, then what?" - if the hypothesized Cause C is *not* present, then what?

1. Event E still occurs: then C doesn't causally determine E
2. Event E does not occur: then C *does* causally determine E

Event: I arrive at the lecture in time.
C: I set my alarm clock
not-C: I don't set my alarm clock - will I make it?

Counterfactuals ☂️

Instead of hypothesized Cause C, we imagine a counterfactual reality with a different C'.

Actual events: I didn't bring my umbrella. It rained. 🌧️
Hypothesized causal relationship: C: `forgetting umbrella` *caused Event* `rain`.
Hypothesized counterfactual C': I did bring my umbrella.
Test: would it still `rain` ?
If `forgetting umbrella` caused `rain`, then `NOT forgetting umbrella` should result in `NO rain`.
If it still `rains` despite having umbrella, then `rain` was not caused by `forgetting umbrella`.

Causal and counterfactual explanations

Were any of your explanations *causal*?

- "Leaves fall in autumn because they get cold and fall off"

Were any of your explanations *counterfactual*?

- "If it wasn't cold in autumn, the leaves wouldn't fall"

Counterfactual Explanations

Explain a certain model decision by showing how a *minimal change in the input* (counterfactual) leads to a different prediction.

Loan Decision Example > Ib gets denied a loan by the bank.

Counterfactual: "If your salary was 6000 dkk higher, the algorithm would have recommended the loan."

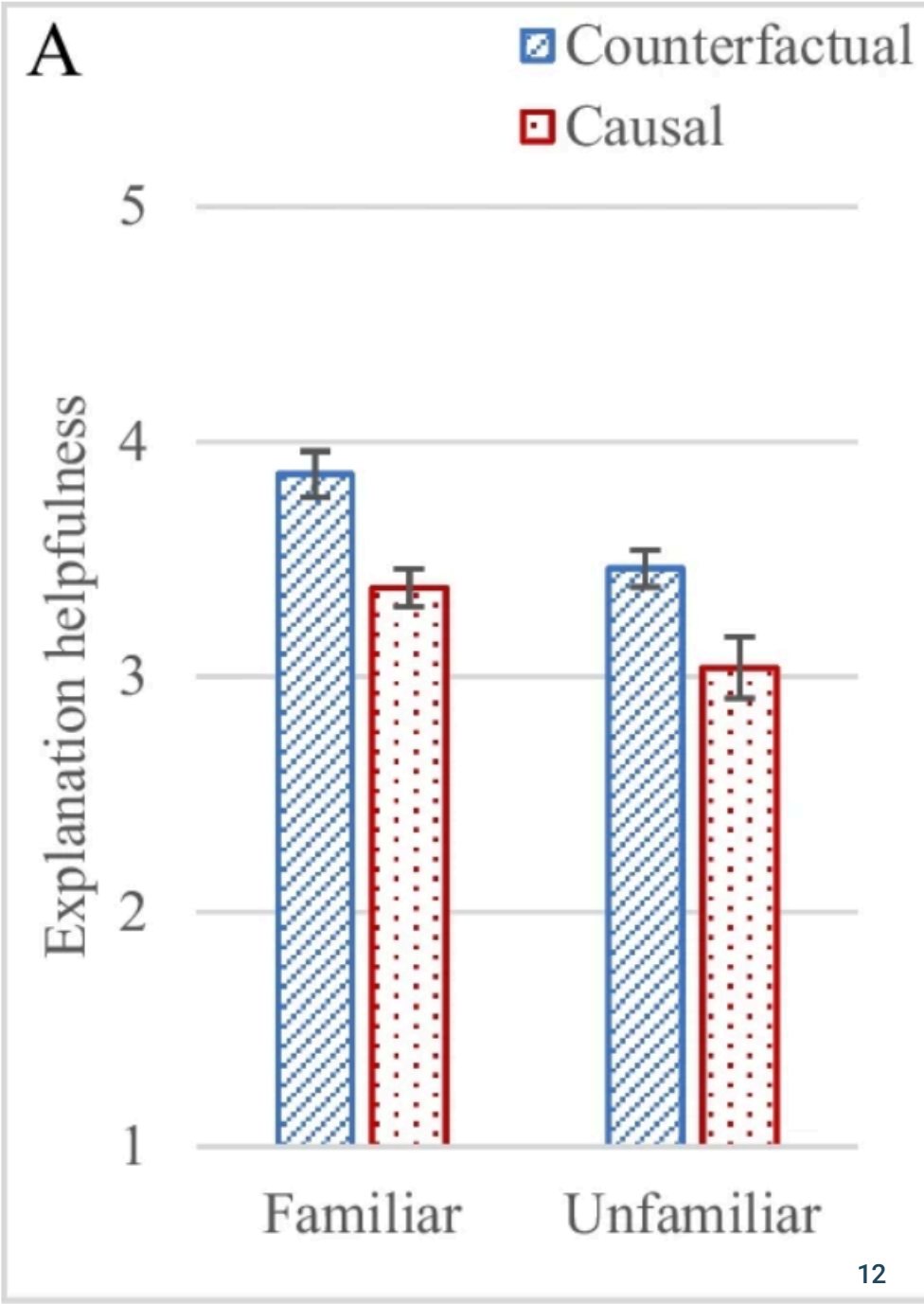
Causal: "Your salary was too low to recommend the loan."

Counterfactual Explanation

Counterfactual explanations seem to be better than causal explanations.

- Judged more helpful by users
- For AI-supported decision making: better than no explanations, same as causal explanations

Celar, L., Byrne, R.M.J. 2023. How people reason with counterfactual and causal explanations for Artificial Intelligence decisions in familiar and unfamiliar domains.



Counterfactual Explanations

How to find good counterfactuals?

Desiderata: They should

- cause a different result / effect E (or minimally, change outcome probabilities signficantly)
- be as similar as possible to the original cause C
- be probable
- be actionable (context-dependent)

Roshomon effect: Multiple different counterfactual explanations could all be valid.

Counterfactual: Crossing the Decision Boundary

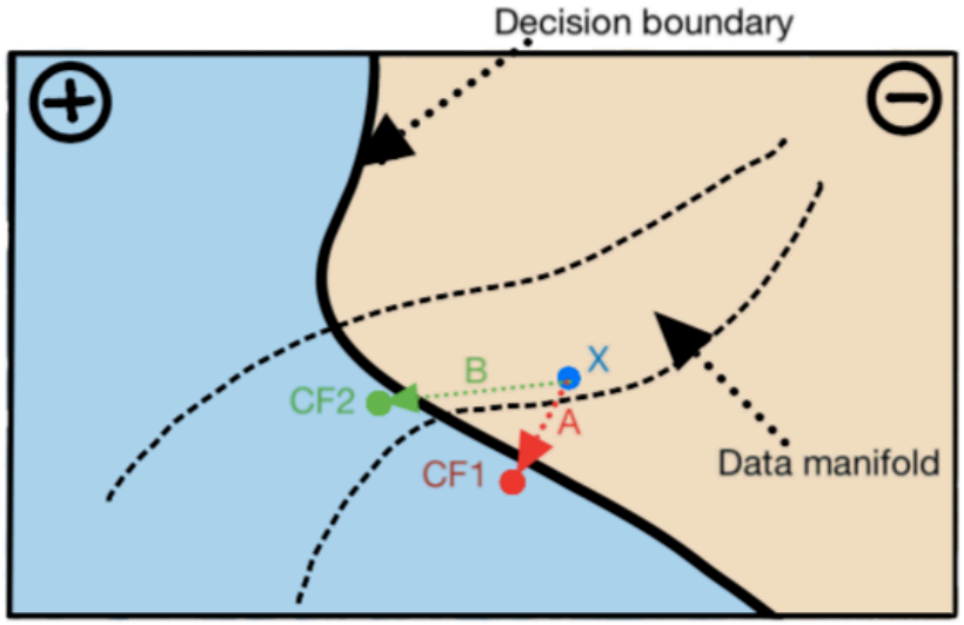


Figure 1: Two possible paths for a datapoint (shown in blue), originally classified in the negative class, to cross the decision boundary. The end points of both the paths (shown in red and green) are valid counterfactuals for the original point. Note that the red path is the shortest, whereas the green path adheres closely to the manifold of the training data, but is longer.

Min-Edit: for x, find a counterfactual x' with desired outcome y' by optimizing loss:

$$\arg\min_{x'} \max_{\lambda} \lambda(f(x') - y')^2 + d(x, x')$$

- minimize distance to target outcome
- minimize distance to original point

Can add additional objectives (e.g. different distances, such as L1 for sparsity)

Verma et al., 2022. Counterfactual Explanations and Algorithmic Recourses for Machine Learning: A Review; Dandl & Molnar, Counterfactual Explanations, Interpretable ML Book

Counterfactual objectives (bonus)

How to reach desiderata: loss function with multiple objectives, e.g.:

$$\arg\min_{x'} \max_{\lambda} \lambda(f(x') - y')^2 + d(x, x')$$

- minimize distance to target outcome: cause the desired (different) result
- minimize distance to original point: be as similar as possible to the original
- sparse changes (L1): change as few attributes/dimensions as possible
- follow data manifold: be probable

See Dandl & Molnar, Counterfactual Explanations, Interpretable ML Book for technical details.

Are CF explanations good explanations? Discuss

- **faithfulness** to model
- **selectivity**: minimal explanation of a single phenomenon at a time
- **sparsity**: doesn't involve more of the model than necessary
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Explanations should be useful for communication:

- human **comprehensibility**
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Mueller et al., 2026: The Quest for the Right Mediator: Surveying Mechanistic Interpretability for NLP Through the Lens of Causal Mediation Analysis

Back to (Causal) Interpretability

- **What** is causing specific behavior?
- **Where** is this behavior localized? Can we look into the "black box"?

Causal/Counterfactual intervention: Would the model behave differently given a small change?

Causal Mechanisms and Mechanistic Intepretability (Mueller++2026)

"In the causality literature, a *mechanism* is defined as a causal chain from cause X to effect Y (Salmon 1984; Pearl 2009)..

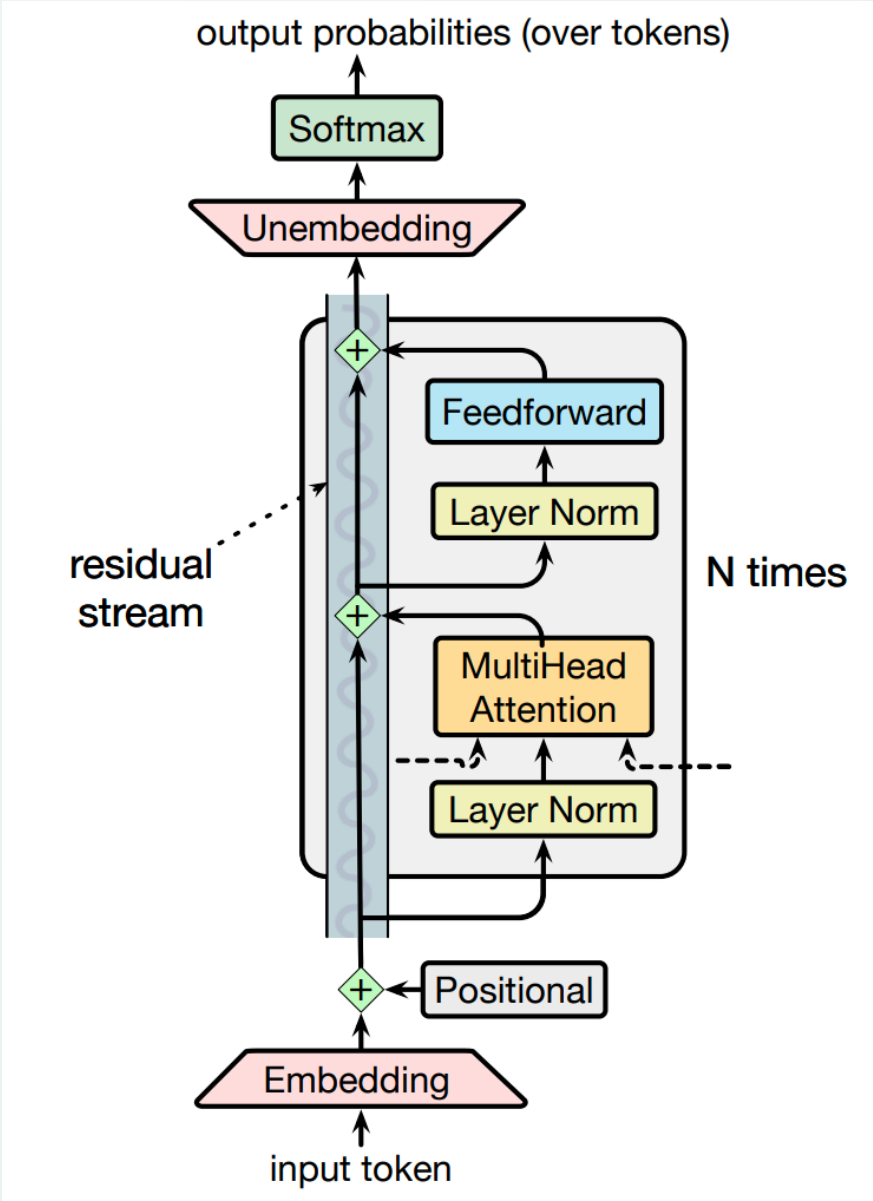
We believe causal definition of “mechanism” is an actionable one that makes the main challenge of mechanistic interpretability more precise—to reverse-engineer algorithmic understanding or control of model behaviors, where “algorithm” is equivalent to a task-specific causal graph H explaining how the model C performs a given task."

What could we change?

Where could we intervene, inside a transformer-based LLM (e.g.)?

What would these interventions tell us?

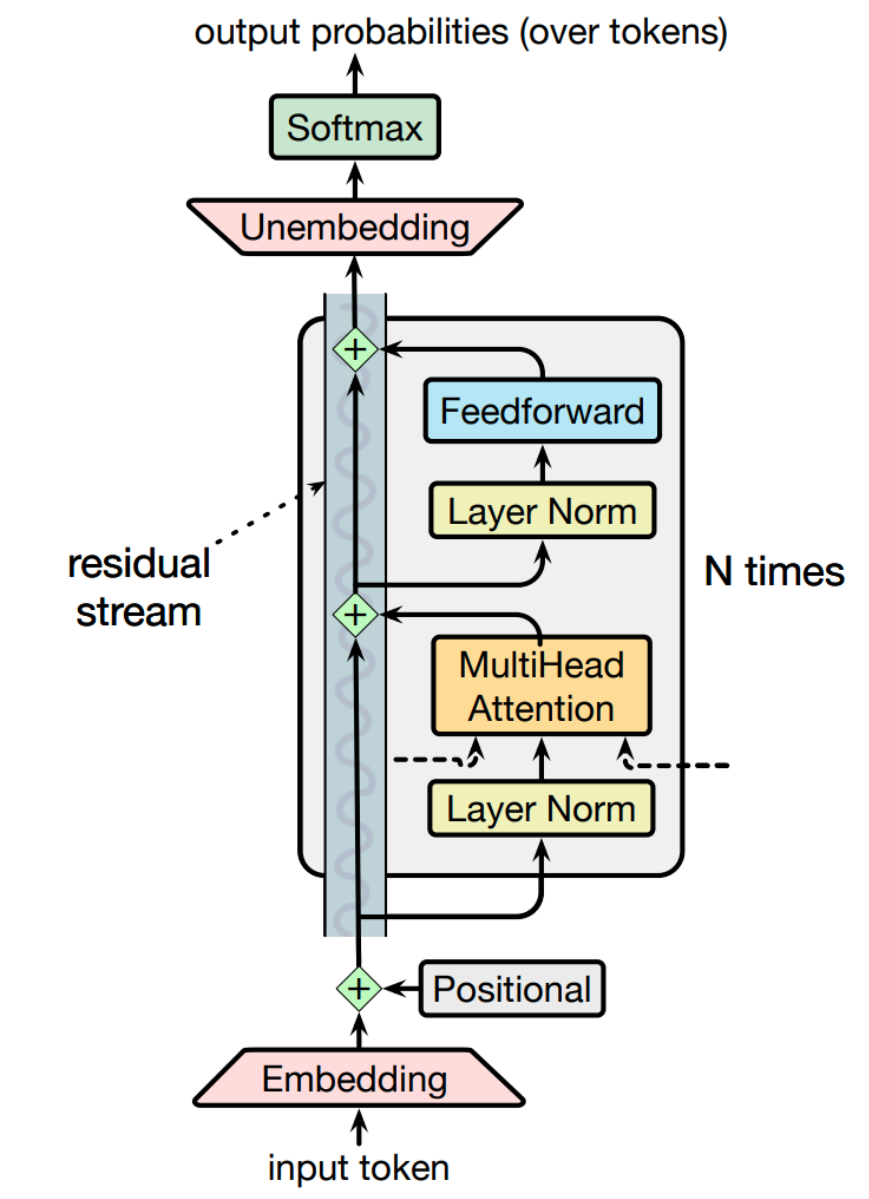
Brainstorm in small groups - 3 minutes.

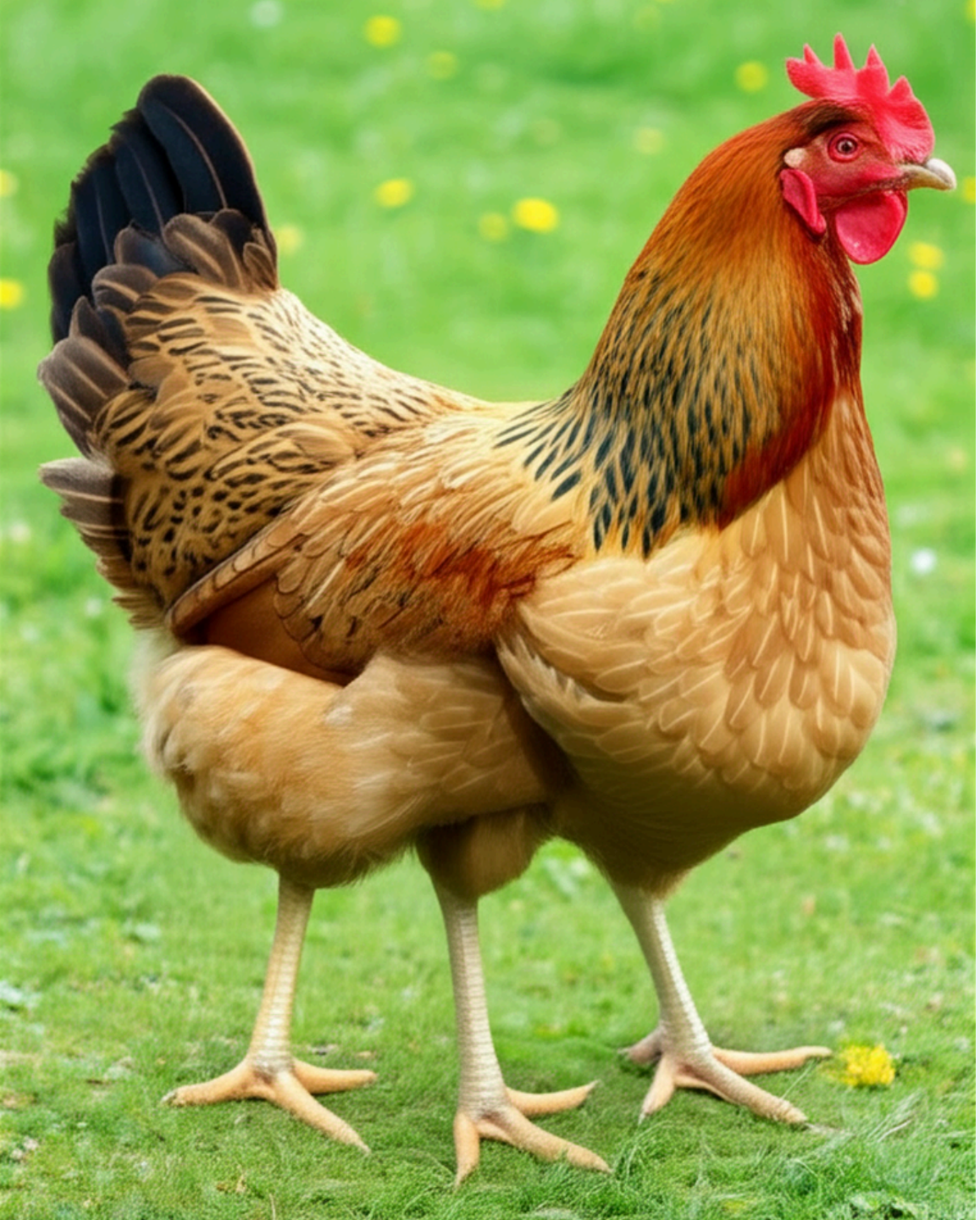


Where could we intervene?

- Inputs
- Attention patterns, attention heads
- Individual neurons
- Intermediate representations

Mueller et al., 2026: The Quest for the Right Mediator: Surveying Mechanistic Interpretability for NLP Through the Lens of Causal Mediation Analysis





Counterfactual Inputs

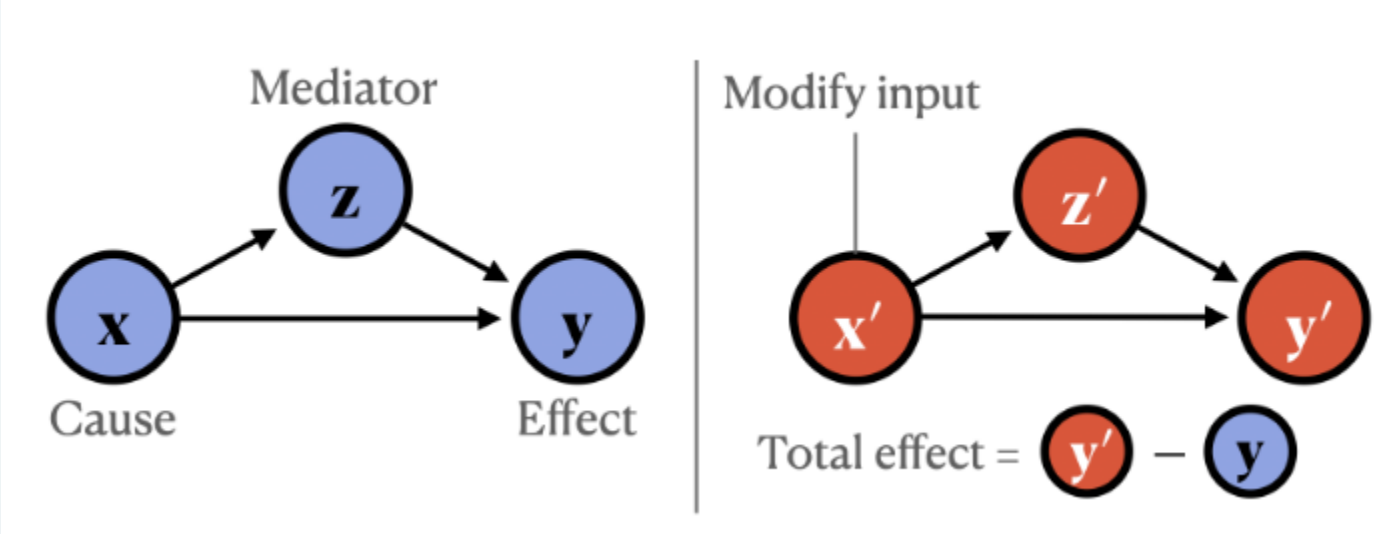
How should we construct CF inputs?

- omit input features
  (e.g. SHAP feature importance)
- change input features minimally, counterfactually

See Project 3 Part 2!

Vo et al., 2025. Vision Language Models are Biased

Causal Graphs and Interventions



(Mueller++26, Figure 2)

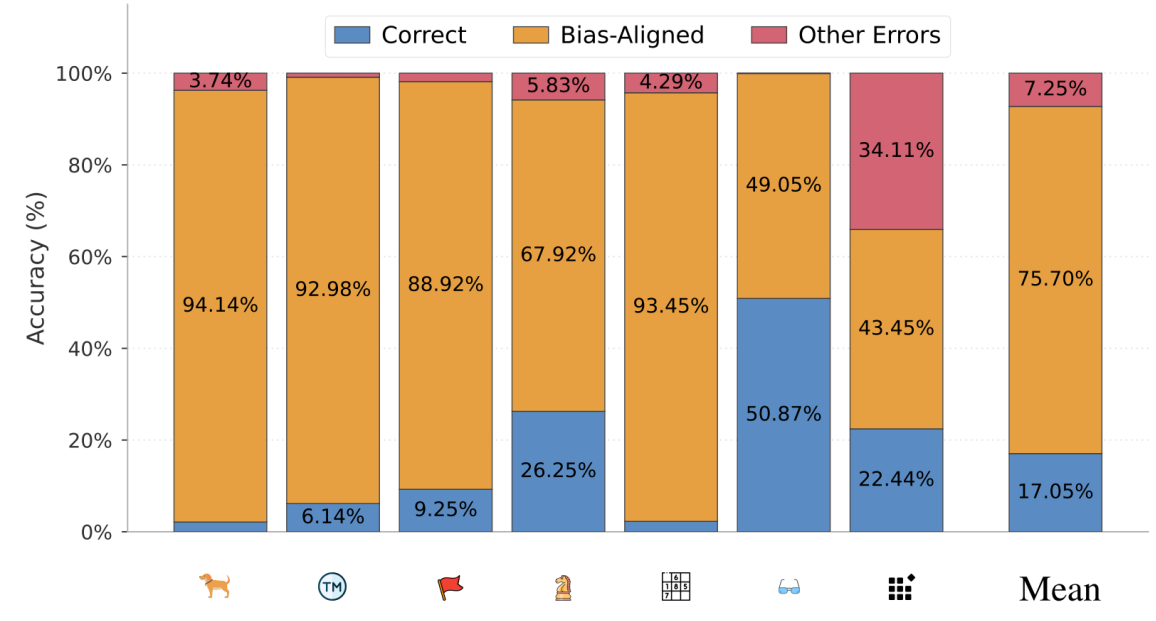
Vo++25. Vision Language Models are Biased

How many legs does this animal have? Answer with a number in curly brackets, e.g., {9}.

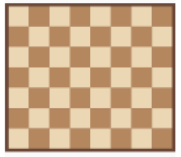
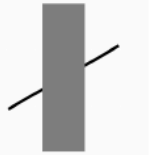
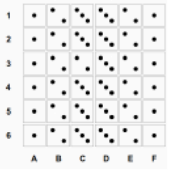
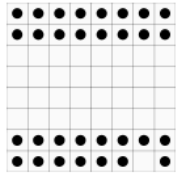
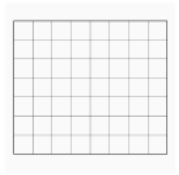
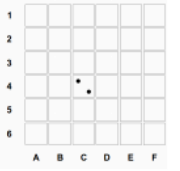
(a)-(e) How many legs does this animal have? Answer with a number in curly brackets, e.g., {9}.

	(a) Lion		(b) Stork		(c) Elephant		(d) Duck		(e) Dog	
Gemini-2.5 Pro	4	✗	2	✗	4	✗	2	✗	4	✗
Sonnet-3.7	4	✗	2	✗	4	✗	2	✗	4	✗
GPT-4.1	5	✓	2	✗	6	✗	2	✗	6	✗
o3	4	✗	2	✗	4	✗	2	✗	6	✗
o4-mini	4	✗	2	✗	4	✗	2	✗	4	✗
Bias	4	✗	2	✗	4	✗	2	✗	4	✗
GT	5	✓	3	✓	5	✓	3	✓	5	✓

Counterfactual images do *not* have an effect - even though they should.



Vo++25. Vision Language Models are Biased

Task	a. Animals	b. Logos	c. Flags	d. Chess Pieces	e. Boardgames	f. Illusion	g. Grids
Approach	LangSAM; cropping	LangSAM	LLM generated SVG code	Script	Script	Script	Script
Before							
After							

Model	Accuracy ↑		Bias rate ↓	
	Before	After	Before	After
Gemini-2.5 Pro	16.02	40.73 (+24.71)	76.79	39.99 (−36.80)
Sonnet-3.7	16.59	42.54 (+25.95)	76.63	39.74 (−36.89)
GPT-4.1	13.88	39.65 (+25.77)	76.62	32.74 (−43.88)
o3	18.50	35.25 (+16.75)	74.81	34.64 (−40.17)
o4-mini	20.25	32.54 (+12.29)	73.66	28.47 (−45.19)
Mean	**17.05**	**38.14 (+21.09)**	**75.70**	**35.12 (−40.58)**

Table 3. Counting accuracy and bias rate (%) before and after removing the background.

Break!

Project 2 Stereotype Leaderboard

Rank	Group	Stereotype hypothesis	Blocks	Qwen bias ×100 ↑	SmolLM2 bias ×100	Qwen UNKNOWN	Qwen disamb. accuracy
1	18	Stuttering / anxiety	16	+50.0	+18.8	6.2%	96.9%
2	5	Hobbies and performance	9	+22.2	-5.6	11.1%	100.0%
3	7	Homeschooling / sociability	16	+18.8	+15.6	43.8%	100.0%
4	9	Remote work / performance	16	+15.6	-3.1	84.4%	87.5%
5	2	Jutland / Copenhagen	12	+12.5	+16.7	54.2%	100.0%
6	6	Mexican / drug trafficking	25	+10.0	+10.0	62.0%	100.0%
7	4	Urban / rural associations	16	+9.4	-3.1	59.4%	84.4%
8	11	High ability / social skills	19	+5.3	-5.3	36.8%	100.0%
9	10	Engineering / social skills	20	+2.5	+2.5	42.5%	100.0%
10	20	Academic field / competence	12	+0.0	-4.2	41.7%	95.8%
11	1	Greek / German work ethic	16	-3.1	+0.0	9.4%	100.0%
12	15	Hobbies / social skills	12	-8.3	-29.2	33.3%	100.0%
13	13 (file 23)	Urban / rural behaviour	12	-16.7	+16.7	50.0%	100.0%

Pathways of Visual Information Flow in Vision-Language Models

Israfel Salazar[1] **Stella Frank**[2] **Dan Oneata**[3,4] **Desmond Elliott**[1] **Constanza Fierro**[1]
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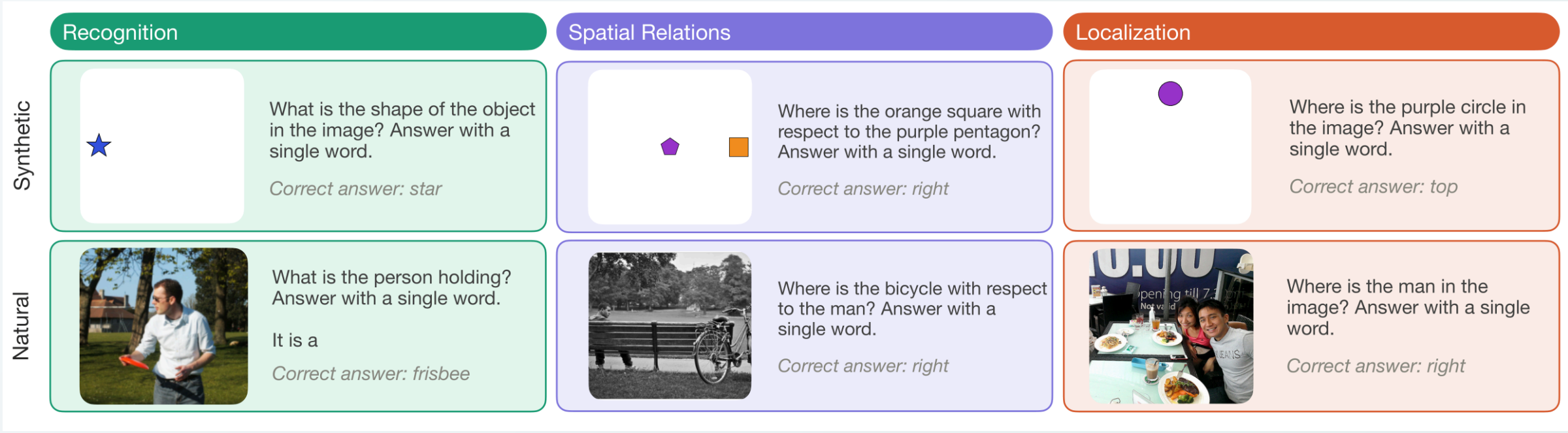
Abstract

We study how visual information is routed in vision-language models (VLMs). Using causal patching on controlled synthetic and natural datasets, we find that models rely on two distinct pathways to solve visual tasks: A *direct pathway*,

Salazar et al., 2026. Pathways of Visual Information Flow in Vision-Language Models

Visual Information Flow (Salazar++26)

Vision-language models need to use information from the image to solve these tasks.



But how do they do it? Where do they do it (localization)? And why do they sometimes fail?

Attention in VLMs

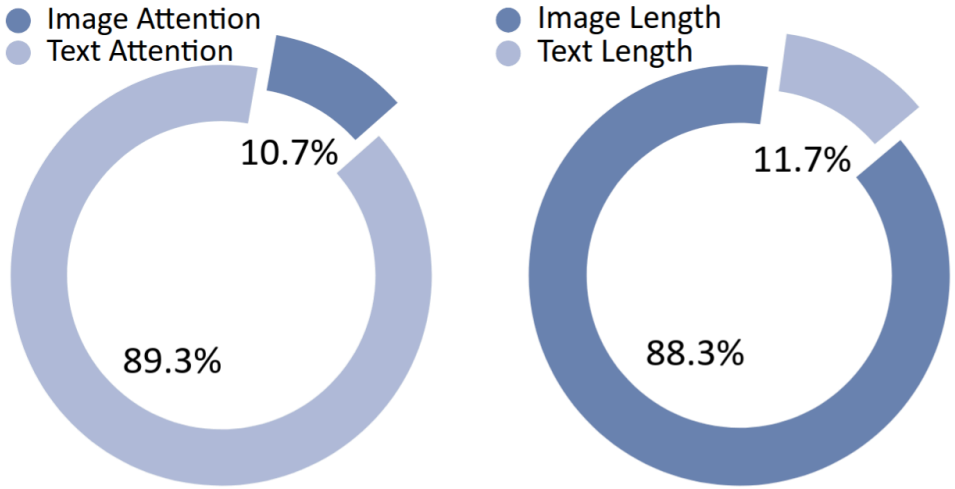
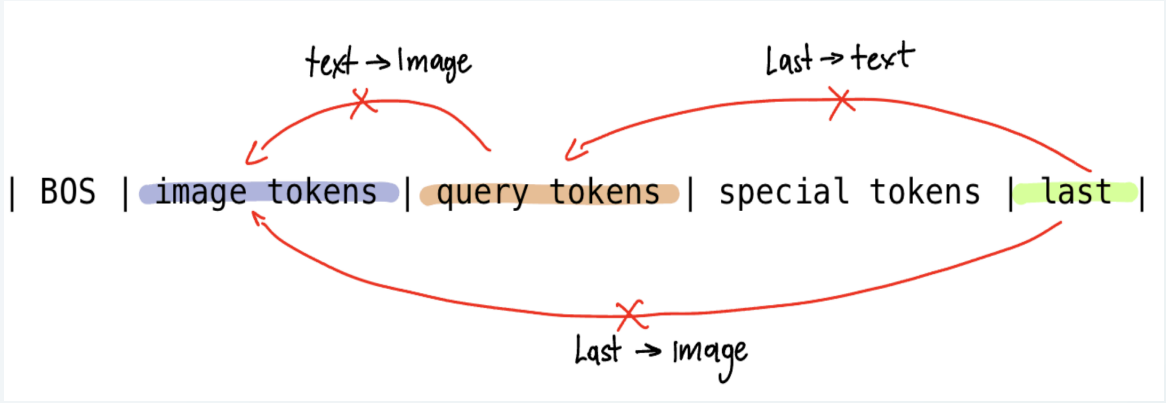


Figure 3. A striking imbalance between visual and textual attention: while image tokens take approximately 90% of the sequence length, they receive only about 10% of the model's total attention on WhatsUp. This severe disparity in attention allocation suggests that VLMs fundamentally underutilize visual information.

Chen et al., 2025. Why Is Spatial Reasoning Hard for VLMs? An Attention Mechanism Perspective on Focus Areas

Salazar++26: Attention knockout



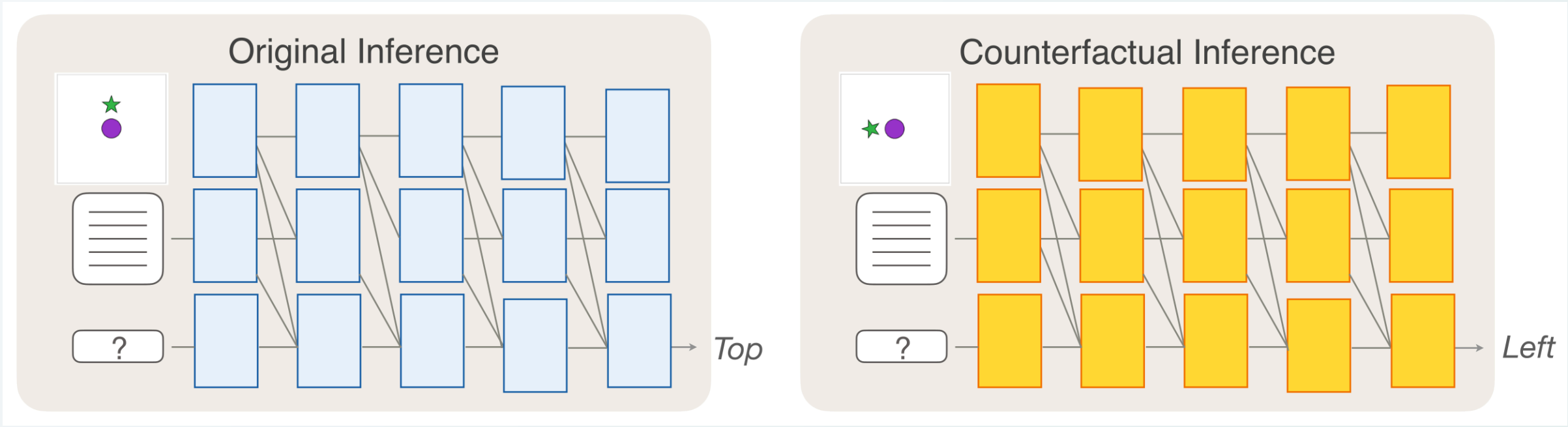
Can a model solve the task without the usual attention pathways?

Change in accuracy after blocking attention:

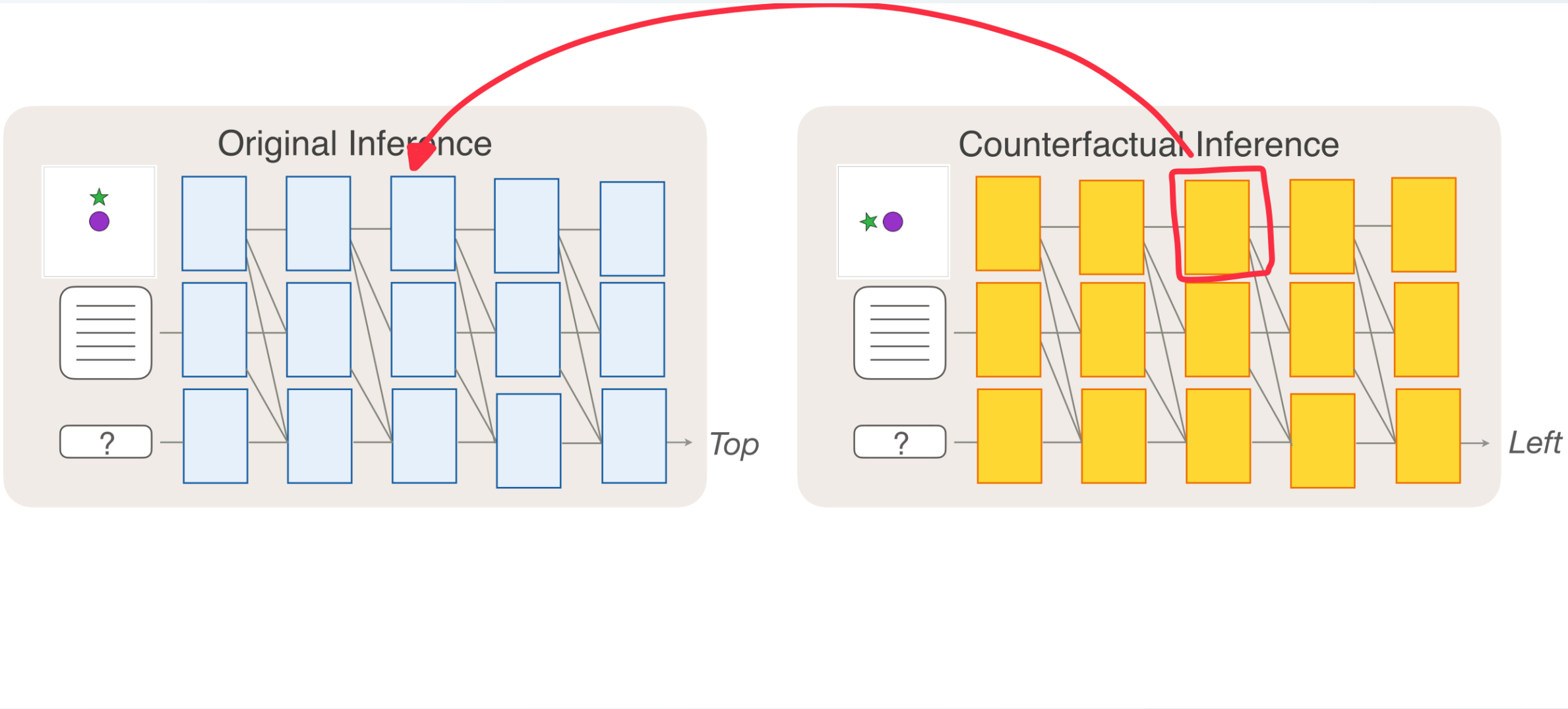
Source	Task	$r_{\mathcal{I}\to\text{last}}$	$r_{\mathcal{I}\to\mathcal{T}}$	$r_{\mathcal{T}\to\text{last}}$
Synthetic	Recognition	0.0	0.0	0.0
	Relations	0.0	-10.5	0.0
	Localization	0.0	0.0	0.0
Natural	Recognition	+1.4	-2.7	+2.1
	Relations	-1.5	-34.9	-1.5
	Localization	+3.6	-18.3	-1.6

Blocking attention doesn't change results. Does this really mean model isn't using attention?

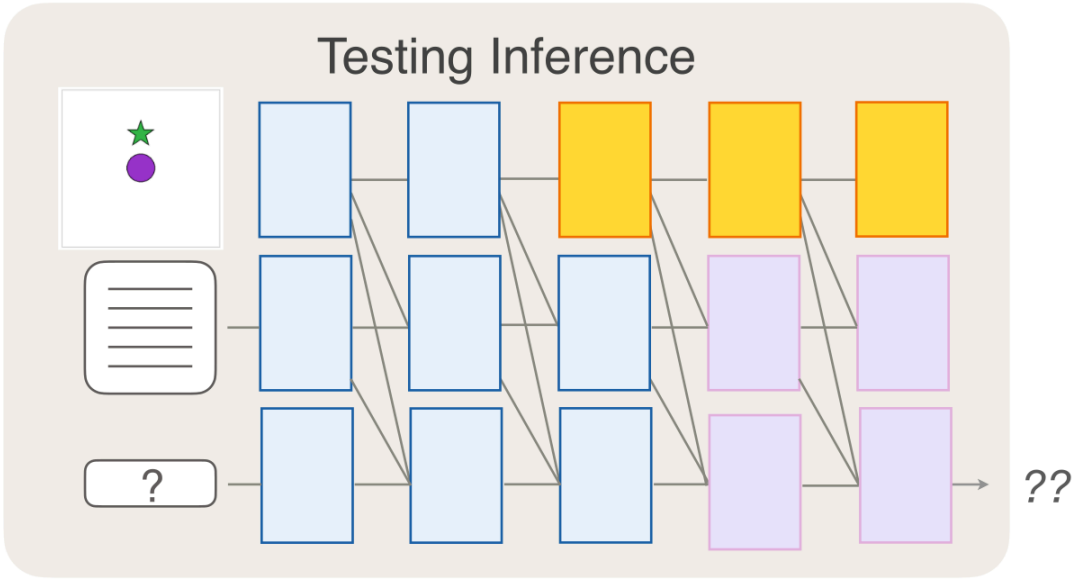
Causal patching



Causal patching



Causal patching



Restoration Score

$$\frac{P_{patched}(\text{“Left”}) - P_{orig}(\text{“Left”})}{P_{CF}(\text{“Left”}) - P_{orig}(\text{“Left”})}$$

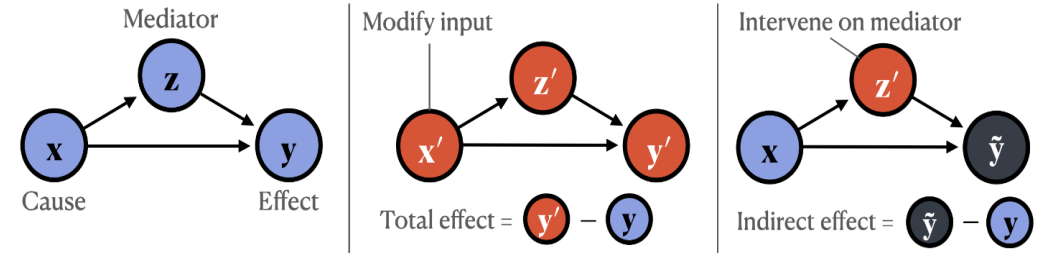
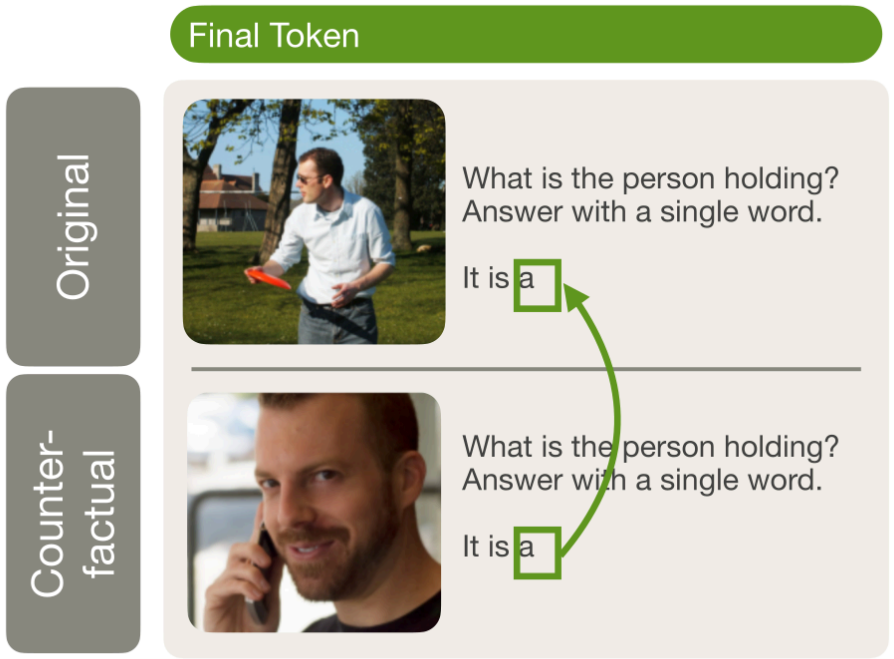
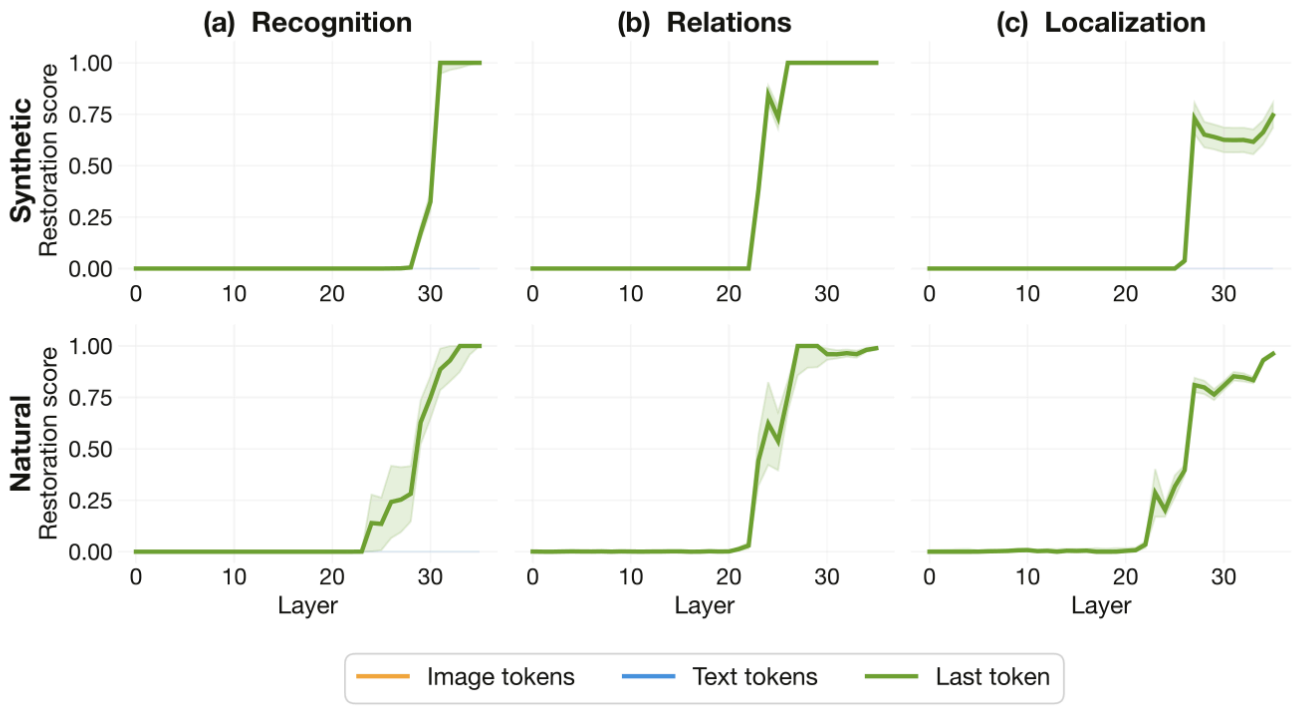


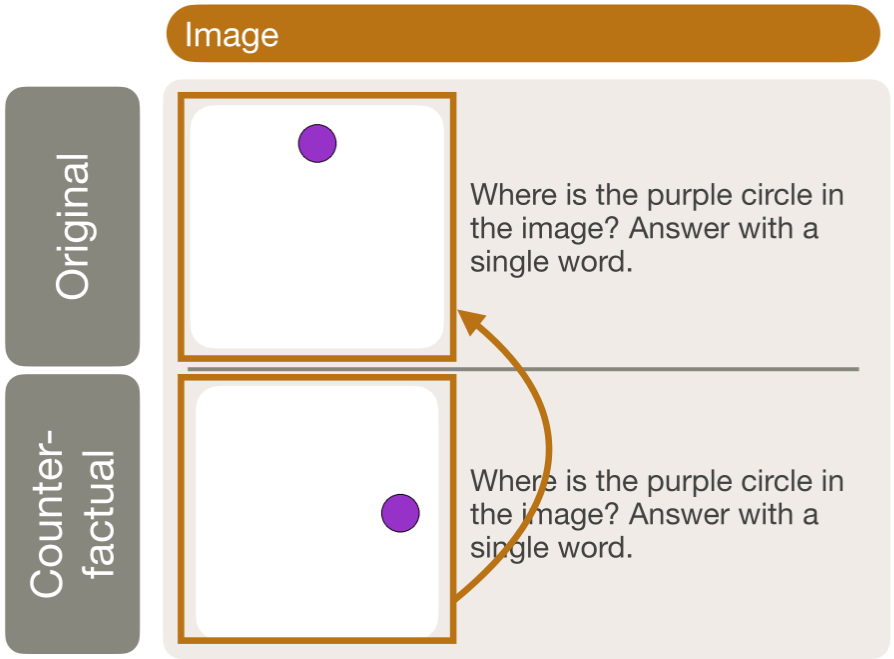
Figure 2
Visual summary of causal mediation analysis. Given input $X = x$ and the resulting output (model prediction) $Y = y$, and another input $X = x'$ that results in different output $Y = y'$, we can compute the total effect of changing x to x' as $y' - y$. In language models, there exist components Z that mediate the influence of X on Y. A common way to quantify the importance of Z is by measuring its **indirect effect** (Equation (1)), where, given $X = x$, one sets Z to some counterfactual value z'. In this figure, we set Z to what it would have been given x'; this results in $Y = \tilde{y}$. One can then measure the indirect effect as $\tilde{y} - y$.

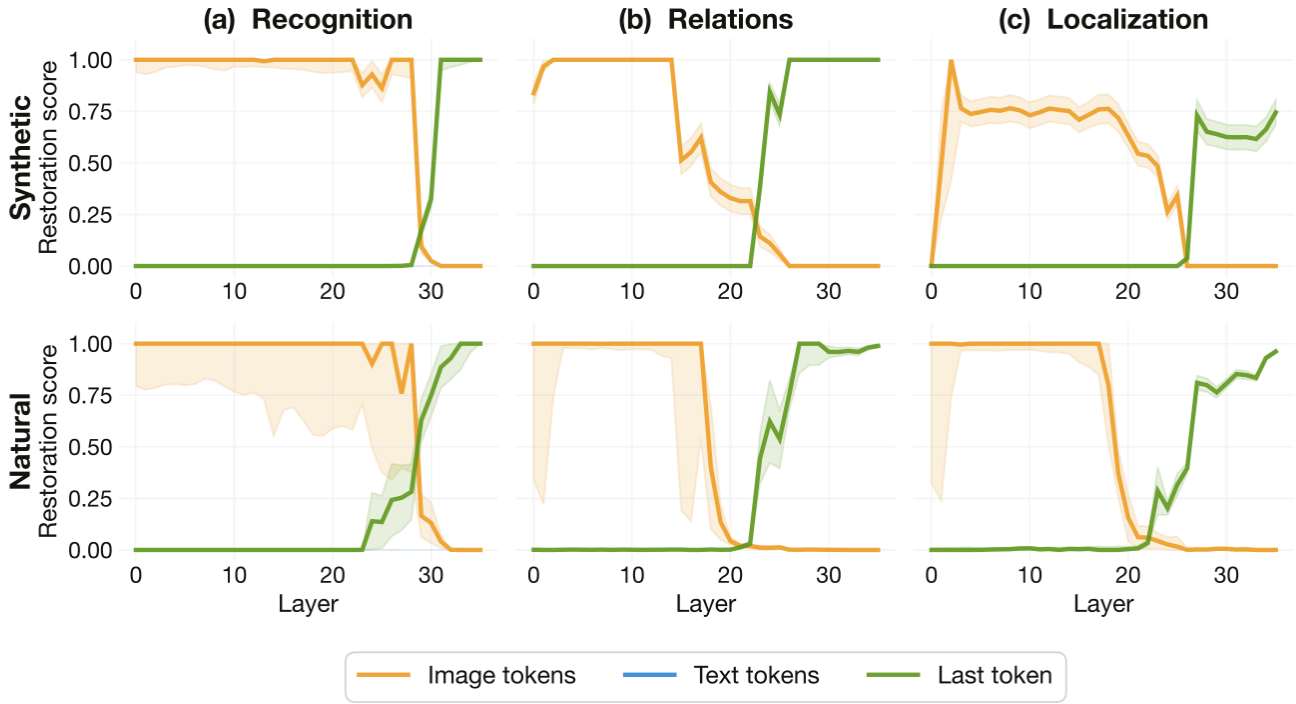
Causal patching: Last Token



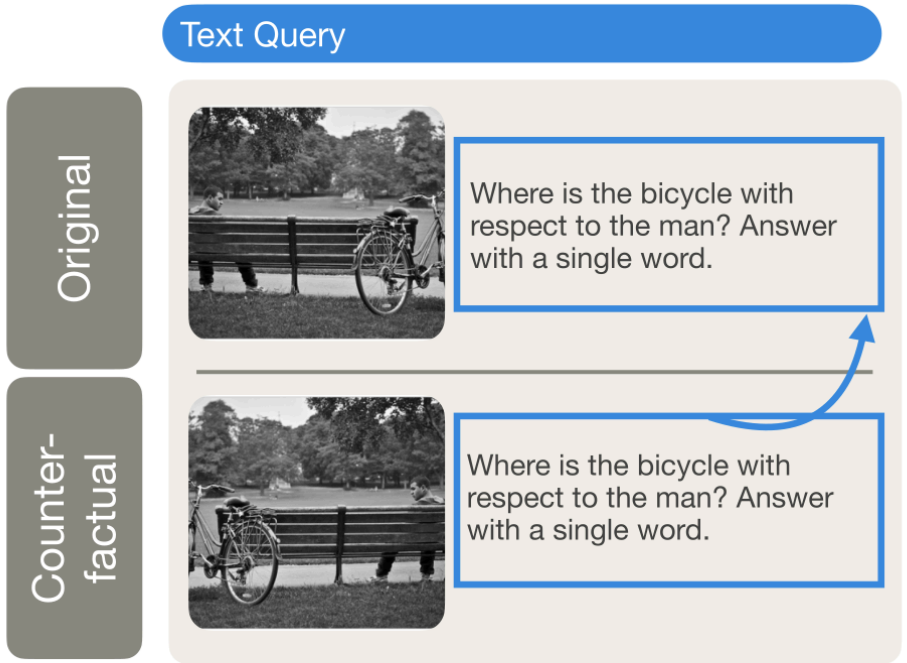


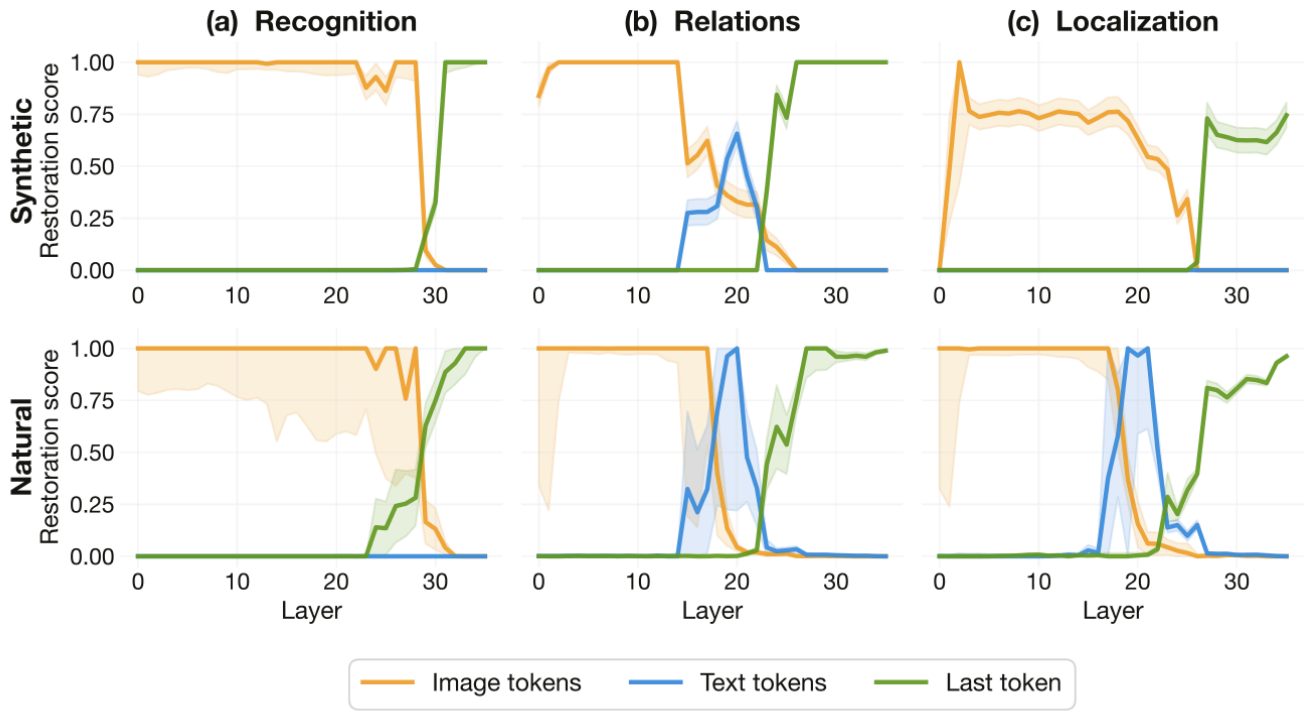
Causal patching: Image



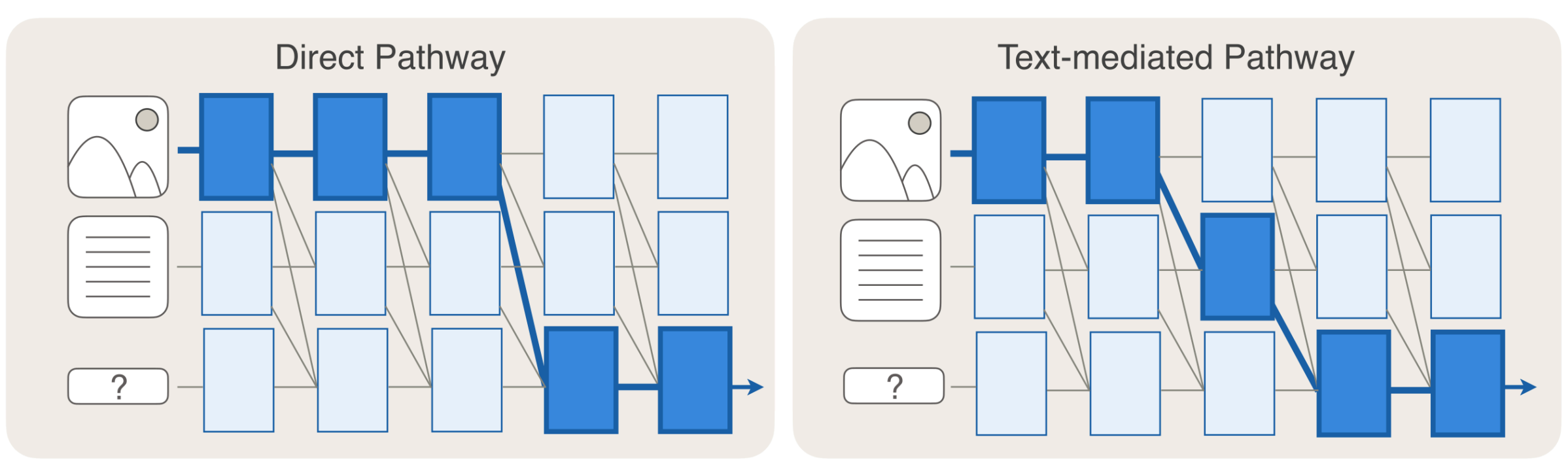


Causal patching: Query

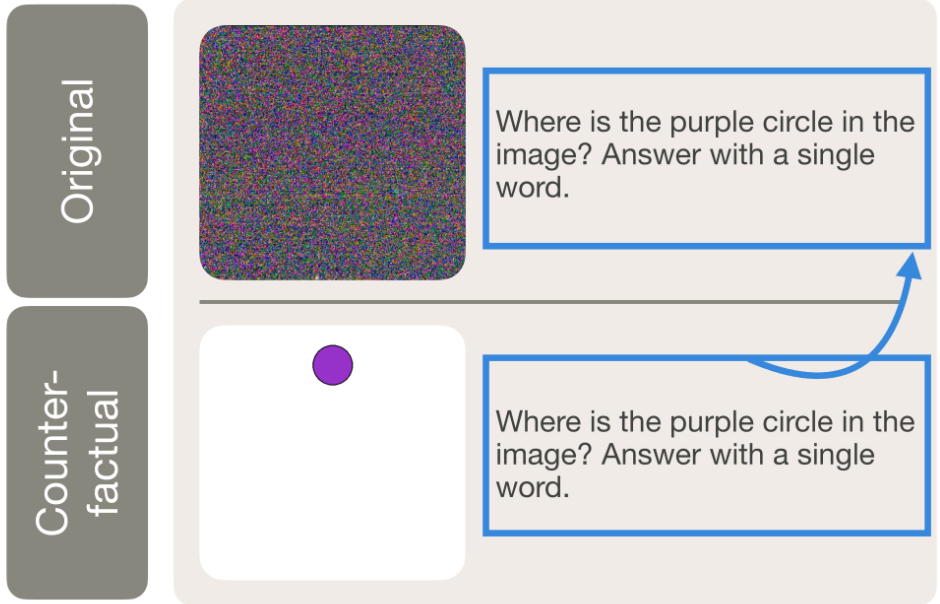




Two Pathways, depending on task



Corrupted patching



Source	Task	Open			Choices		
		N	RS	Acc (%)	N	RS	Acc (%)
Synth.	Recog.	400	1.00	100.0	400	1.00	100.0
	Relat.	400	1.00	100.0	400	1.00	100.0
	Local.	400	0.08	100.0	400	0.80	100.0
Nat.	Recog.	144	0.82	81.9	146	0.99	99.3
	Relat.	1361	1.00	94.9	1401	0.98	97.9
	Local.	1897	1.00	95.8	2471	0.92	92.7

Text from CF is enough to recover the CF answer, even for direct-pathway tasks (object recognition).

- In normal paired causal patching, both pathways carry competing information.
- In corrupted patching, only text carries information: text 'wins', rather than task failure.

Questions?

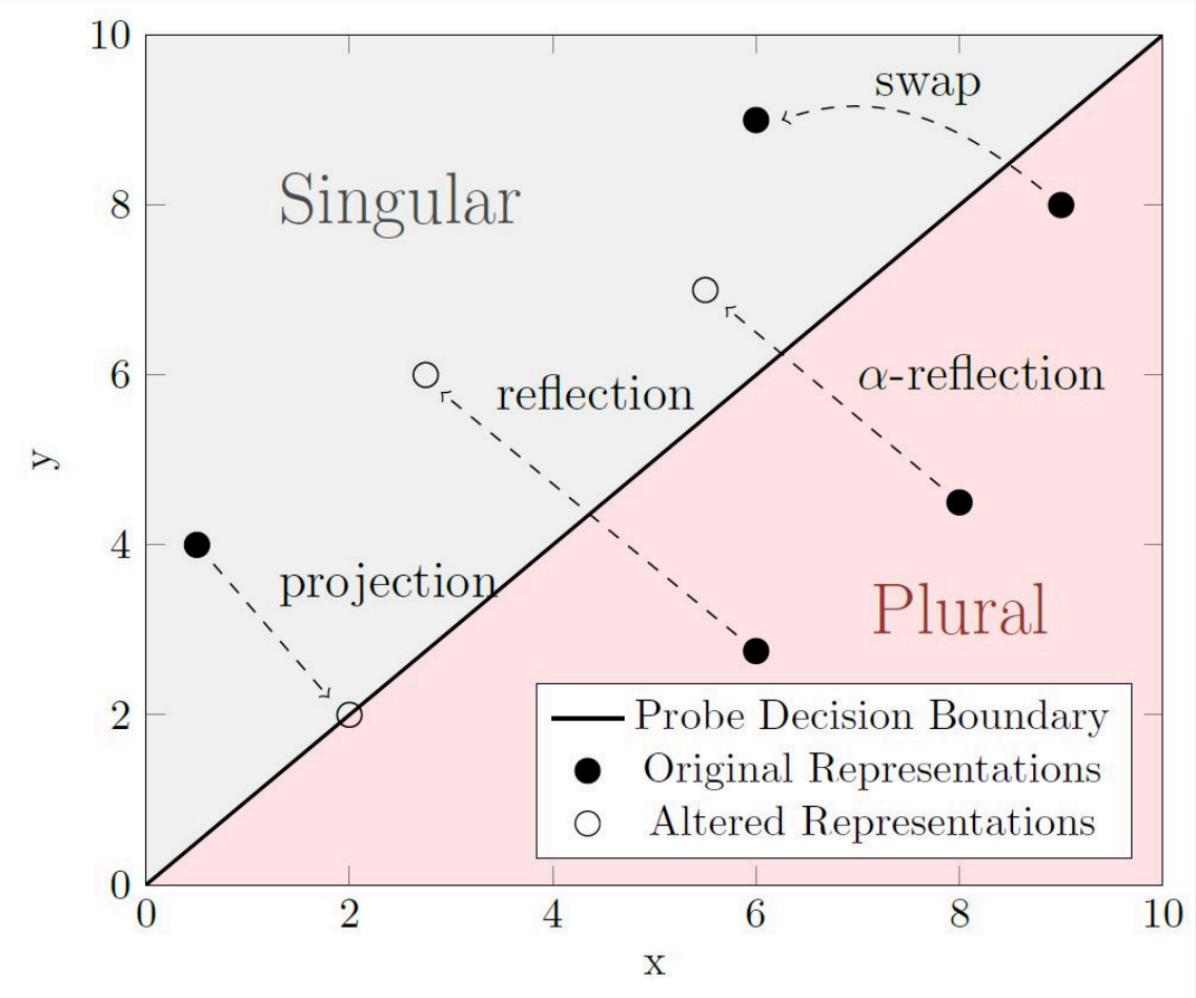
Today

- Causal methods using counterfactuals
- Counterfactual explanations
- Counterfactual model interpretability
  - input interventions, activation patching
  - did not look at: causal versions of linear probing

Project 3 Part 2 is out!

https://github.com/AYAELGEBALY/responsible-ai-rexgradient/tree/main/week2

Linear Probing Intervention



Hanna, 2022 - See also steering vectors.